

Chapter 3: Matchings

3.1 Matchings and Perfect Matchings

A *matching* in a graph is a set of edges no two of which are adjacent. A vertex is *saturated* if it is the end of an edge in the matching, and a *perfect matching* (sometimes called a 1-factor) saturates all the vertices.

EXAMPLE. Consider counting the perfect matchings of the complete graph K_{2n} . One way to proceed is to choose an ordering of the vertices—in $(2n)!$ ways—and build a perfect matching by pairing first and second, third and fourth, and so on. But since which vertex is first and which is second doesn't matter, and the order of the edges doesn't matter, any particular matching is built in $n!2^n$ different ways. Another way to proceed is to choose the partner for the smallest numbered vertex, in $2n - 1$ ways. Then the partner for the smallest unmatched vertex can be chosen in $2n - 3$ ways. And so on. Thus the number of perfect matchings is

$$\frac{2n}{2^n n!} = (2n - 1)(2n - 3) \cdots 1$$

(This quantity is sometime written as $(2n)!!$ with the double factorial.)

In general we care about the maximum size of a matching. Note that a maximal matching is not the same as a maximum matching: the middle edge of P_4 is a maximal matching but not maximum. There is a famous characterization of when a matching M is maximum. Given a matching M , an *M -alternating* path is one where the edges alternate between in and out of M . An *M -augmenting* path is a nontrivial M -alternating path that starts and ends at an M -unsaturated vertex.

BERGE. A matching M is maximum if and only if there is no M -augmenting path.

PROOF. Assume first there is an M -augmenting path. Then one can flip membership of the edges on the path, say from out-in-out-in-out to in-out-in-out-in, and so increase the size of the matching by 1. That is, M is not maximum.

Assume second that M is not maximum. For two matchings (or graphs in general) define the ***symmetric difference*** as the subgraph with the edges that are in precisely one of the graphs. The key observation is that, if M_1 and M_2 are matchings, then their symmetric difference consists of paths and even cycles. (It is immediate that the symmetric difference has maximum degree 2 and so consists of paths and cycles; but the edges alternate being in M_1 and M_2 so only even cycles are possible.)

Now, let M^* be a maximum matching and consider the symmetric difference S of M and M^* . Since M^* has more edges than M in G , this also holds in S . So there is a component of S where M^* is in the majority. This must be an odd path, starting and finishing with an M^* edge, and by the way S is constructed, the ends of the path are not M -saturated. That is, the path is M -augmenting. $\square$

Berge's theorem gives an algorithm to find a maximum matching in a bipartite graph. Starting at an unmatched vertex, one determines the set of vertices reachable by M -alternating paths, and thus either finds an M -augmenting path or proves that none exists.

Another characterization of when a graph has a perfect matching is based on odd components (meaning odd order).

TUTTE'S 1-FACTOR THEOREM. *A graph G has a perfect matching if and only if $o(G - S) \leq |S|$ for all sets of vertices S , where $o(G - S)$ is the number of odd components of $G - S$.*

PROOF SKETCH. For each odd component of $G - S$, one of its vertices needs to be matched with a vertex of S . Thus if $o(G - S) > |S|$ then a perfect matching is not possible. We omit the proof of the reverse direction. $\square$

The following result can be deduced from the previous:

PETERSEN'S THEOREM. *A 3-regular graph without a bridge has a perfect matching.*

3.2 Vertex Covers and Matching in Bipartite Graphs

For a set of vertices X , we let $N(X)$ denote the *neighborhood* of X ; that is, all vertices that have a neighbor in X .

HALL'S THEOREM. *In a bipartite graph with one side X : there is a matching that saturates X if and only if $|N(S)| \geq |S|$ for all $S \subseteq X$.*

PROOF. The condition is obviously necessary, as a saturating matching provides a distinct neighbor for every vertex in X . So we focus on the proof of sufficiency.

Assume $|N(S)| \geq |S|$ for all $S \subseteq X$. Suppose the maximum matching M does not saturate vertex x_0 in X . Then the condition means x_0 has a neighbor, say y_0 . Since the path x_0y_0 is not augmenting, it must be that y_0 is saturated, say with edge to x_1 . Since $|N(\{x_0, x_1\})| \geq 2$, there is a vertex y_1 that is adjacent to one or both of x_0 and x_1 . In either case there is an M -alternating path from x_0 to y_1 ; thus y_1 must be saturated. And so on. We build a set $y_0, y_1, \dots$ each vertex of which is reachable from x_0 by an M -alternating path, and when this number exceeds the number of edges in M , we have an M -augmenting path. $\square$

An immediate corollary is:

REGULAR BIPARTITE. *A regular bipartite graph can be decomposed into perfect matchings.*

PROOF. Let G be a k -regular bipartite graph (with $k > 0$). We observed earlier that the two partite sets have the same size. We claim that it satisfies the condition in Hall's Theorem. Let A be the number of edges between S and $N(S)$. Then $k|S| = A \leq k|N(S)|$, by the degrees. It follows that G has a matching saturating X . Finally note, that such a matching is a perfect matching, and when we remove it, what remains is a $(k - 1)$ -regular bipartite graph. $\square$

A **vertex cover** is a set of vertices intersecting all edges. It is immediate that a set is a vertex cover if and only if its complement is an independent set. An **edge cover** is a set of edges saturating all vertices. Notation: $\alpha(G)$ is the maximum size of an independent set of vertices, $\beta(G)$ is the minimum size of a

vertex cover, $\alpha'(G)$ is the maximum size of a matching, and $\beta'(G)$ is minimum size of edge cover. (Note that we use the primes for the edge versions.)

KÖNIG–EGERVÁRY. *If G is a bipartite graph then $\alpha'(G) = \beta(G)$.*

PROOF SKETCH. A vertex cover must contain one end of every edge in a matching. Thus for all graphs $\beta(G) \geq \alpha'(G)$. To prove equality for a bipartite graph with bipartite sets X and Y , let Q be a minimum vertex cover. Partition the graph into four sets $Q_X = Q \cap X$, $Q_Y = Q \cap Y$, $T_X = X - Q_X$ and $T_Y = Y - Q_Y$.

We claim that in the bipartite graph induced by $Q_X \cup T_Y$ there is a matching that saturates Q_X . For consider $S \subseteq Q_X$ and $N(S) \cap T_Y$. The key observation is that the union of $Q - S$ and $N(S) \cap T_Y$ is a vertex cover of G : all edges that aren't covered by $Q - S$ have an end in S and hence also have an end in $N(S) - Q$. Thus, if $|N(S) \cap T_Y| < |S|$ we have a contradiction of the minimality of Q . Hence we can apply Hall's theorem. Similarly in the graph induced by $Q_Y \cup T_X$ there is a matching that saturates Q_Y . So we have a matching of size $|Q_X| + |Q_Y|$, which is what we were trying to prove. $\square$

It is immediate that for if G is a graph of order n , then $\alpha(G) + \beta(G) = n$. Gallai showed that $\alpha'(G) + \beta'(G) = n$ as well.